



Storytelling; Visual Literacy; Bias;

DSCI 454
Week 2, Lecture 2



What are we learning this lecture?

Communicating Insights to Decision Makers

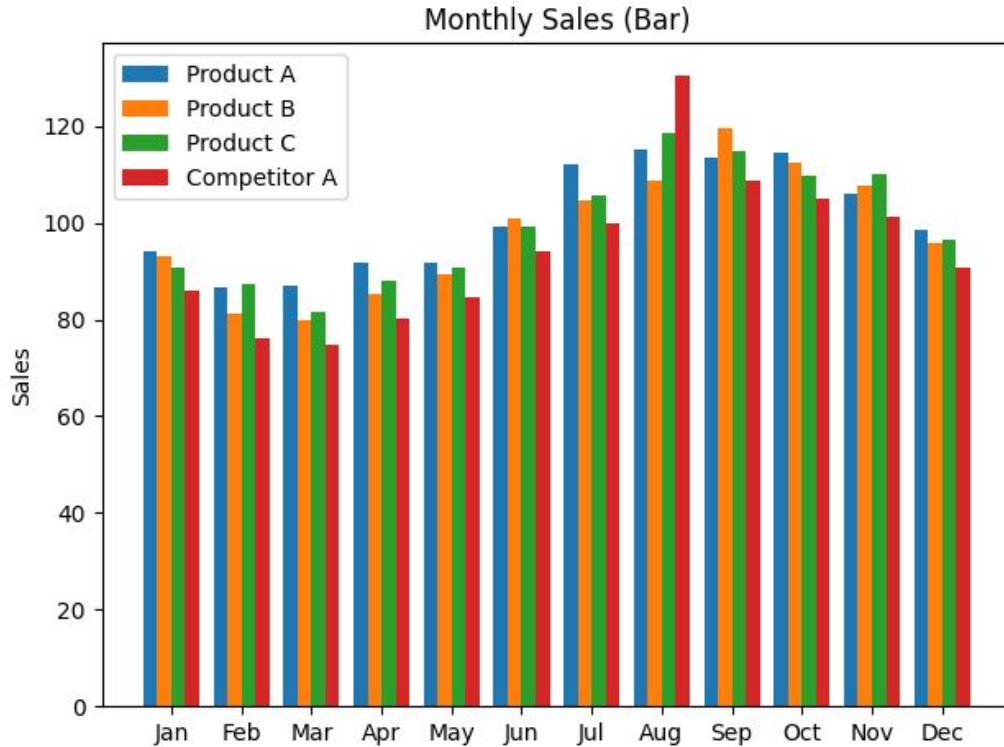
- Storytelling vs reporting.
- Ethical visualization and bias awareness.
- Visual literacy for executives, regulators, and the public.



Storytelling vs. Reporting

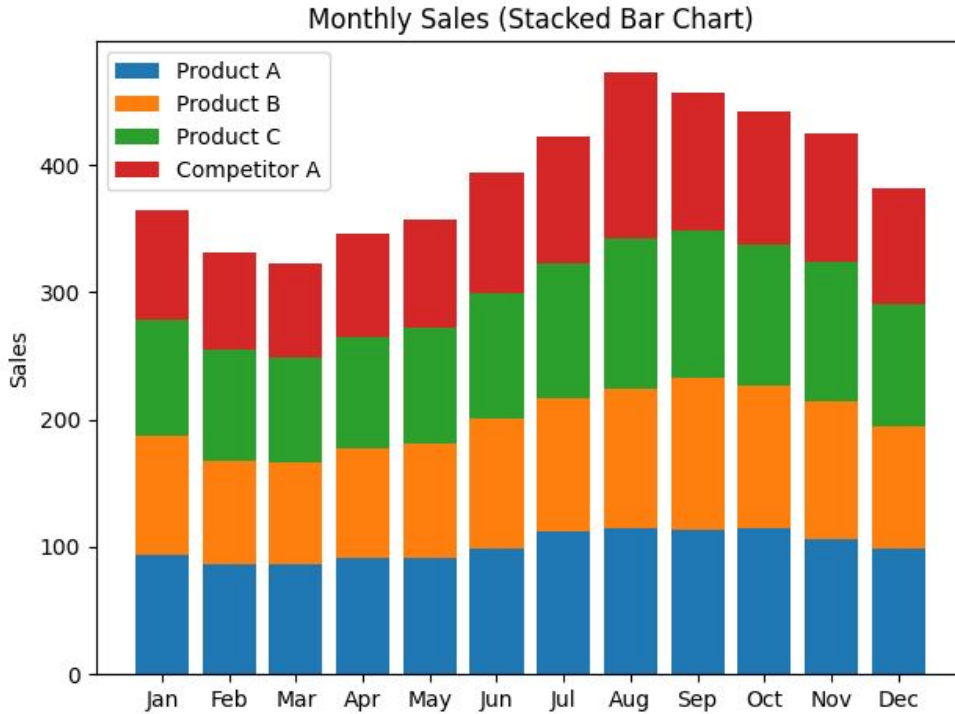
Series adapted from
(*Storytelling w/ Data*, Knaflitz, 2015, p. 187)

Reporting



- “All 100 oysters”
- Similar to exploratory vs explanatory data analysis, this just shows all the raw data, not information.
- Cluttered, overly colorful
- Who’s the Audience?
- What’s the Ask?

Reporting another way

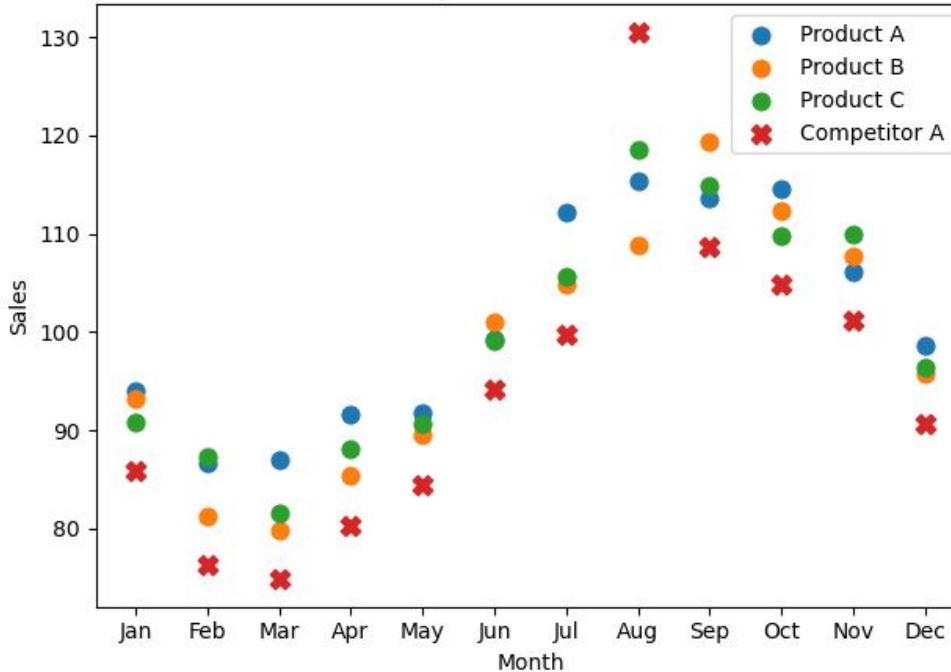


- “100 oysters” another way
- Still just shows all the raw data, not information.
- Still cluttered, overly colorful
- Who’s the Audience?
- What’s the Ask?

Reporting yet another way

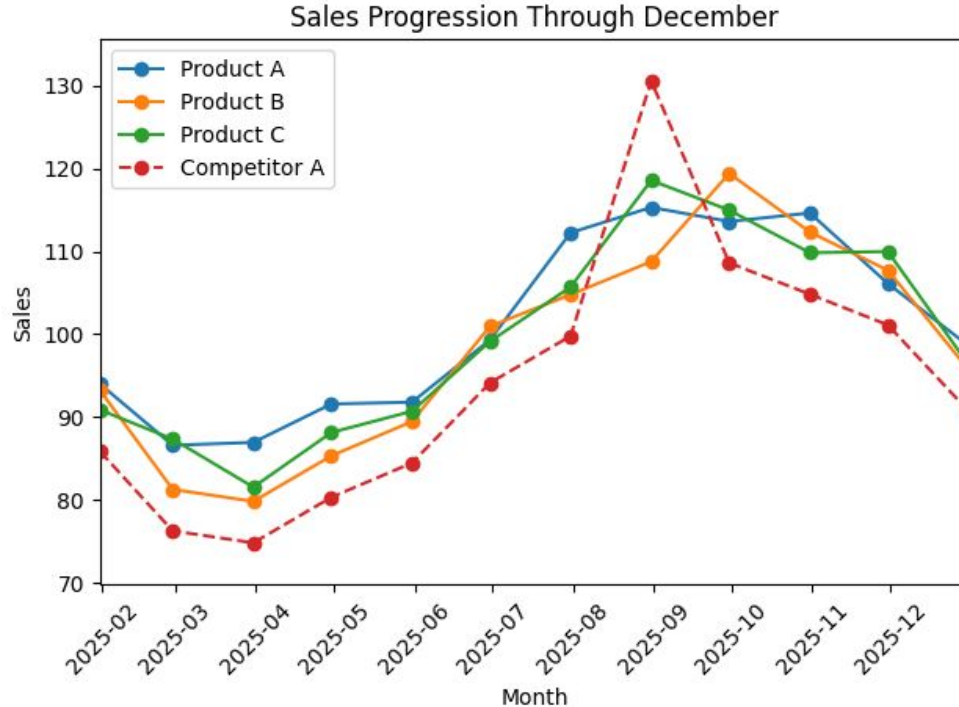


Monthly Sales (Scatter Plot)



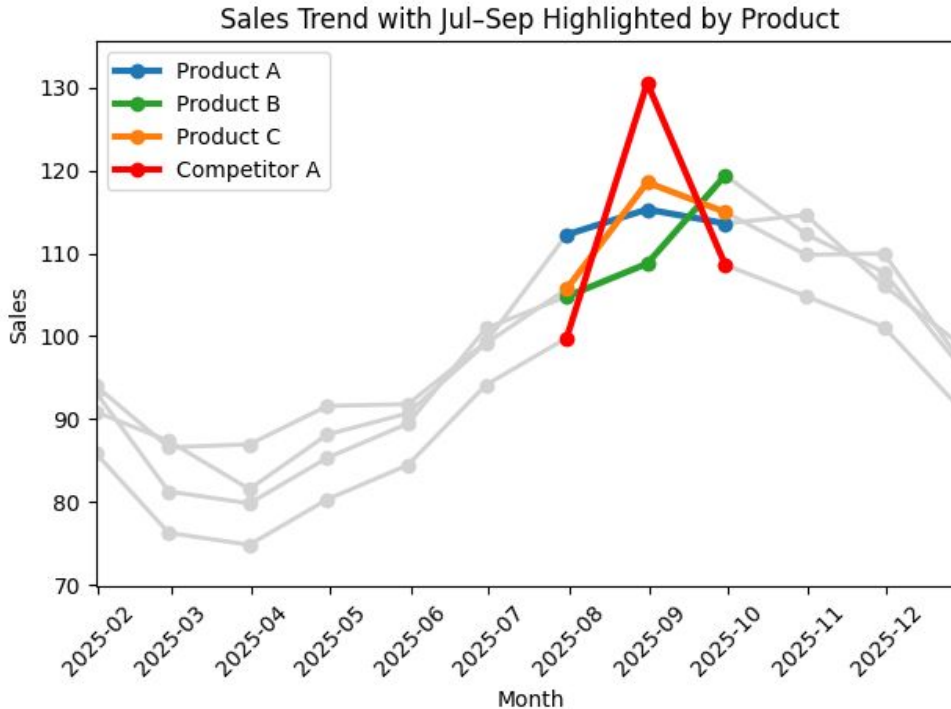
- Better, not best
- Less cluttered, still busy, overly colorful
- Highlights anomaly, outlier in Aug/Sept.
- But still:
 - Who's the Audience?
 - What's the Ask?
- Trends?

Reporting to Storytelling



- Lines are good for time-series data and showing trends, comparisons
- Seasonal trend is clear now
 - Spike in August
 - Back to school?
- But still busy, cluttered, overly colorful

Storytelling



- Declutter
- Highlight what you want your Audience to focus on
 - Neutralize what you don't
 - Only the anomaly is colored
- Focus on the difference between our Product C and Competitor A
- Recommend “Back to School” promotion in Aug.

Attention Check

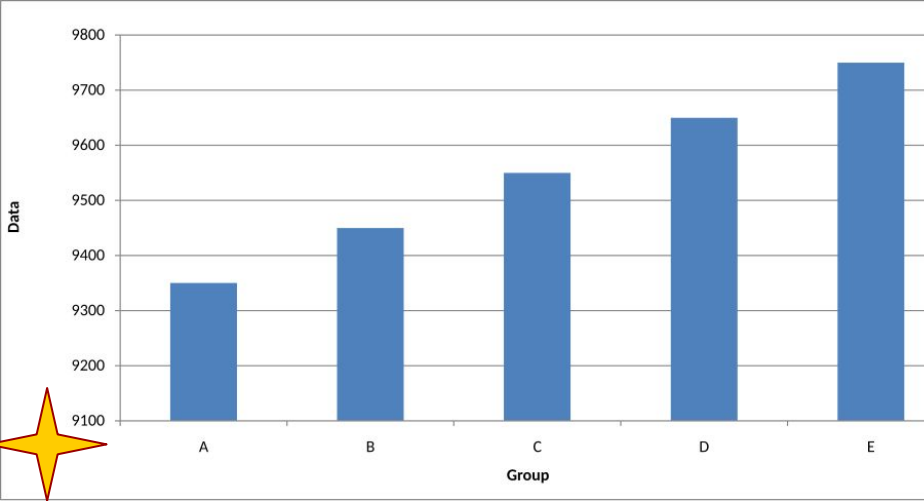


- Did you catch **BOTH (2)** insights from the previous example?
- How could we improve highlighting to show both?

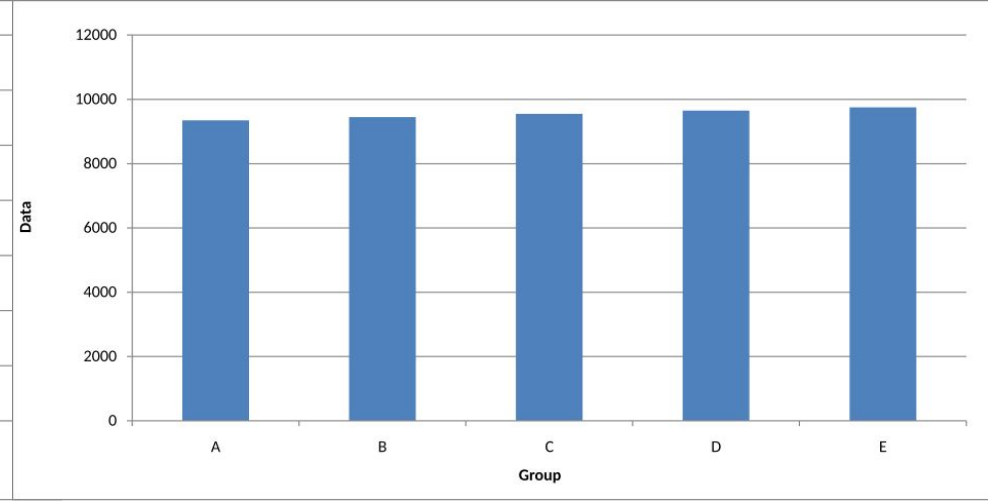


Ethical visualization and bias awareness

Bias: Truncated y-axis



y-axis start: 9100



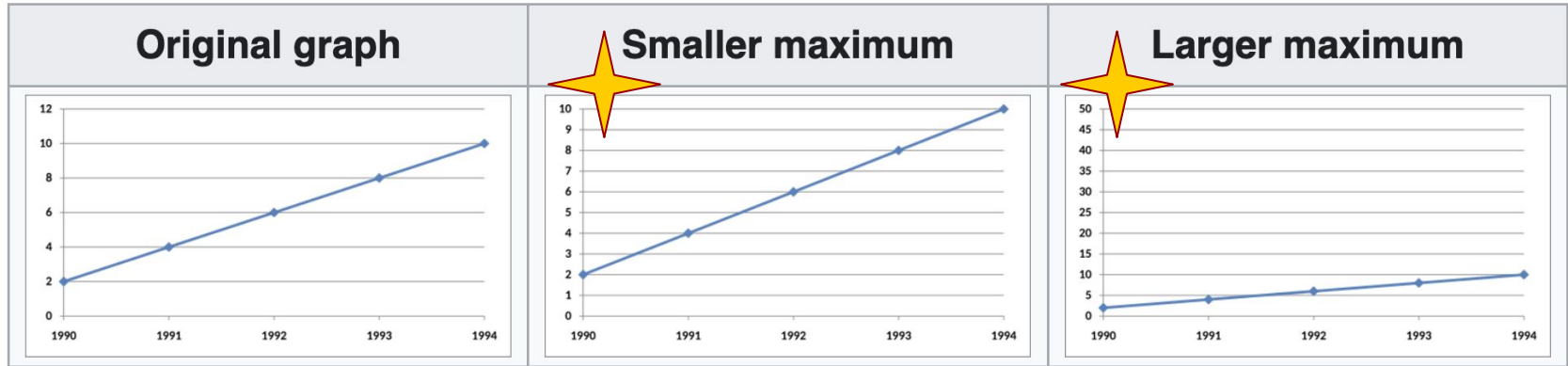
y-axis start: 0

Note: The y-axis does not start at 0. This is often used to exaggerate change.

Bias: y-axis max



Changing y-axis maximum



y-axis max: 12

y-axis max: 10

y-axis max: 50

Note: The y-axis max changes. This can be used to exaggerate or minimize/suppress change.

Bias: Selection / Cherry picking windows



S&P 500

6,940.01 ↓ 0.32% -22.55 5D

Jan 16, 4:44:54 PM UTC-5 · INDEXSP · Disclaimer



5 day window

S&P 500

6,940.01 ↑ 826.94% +6,191.31 MAX

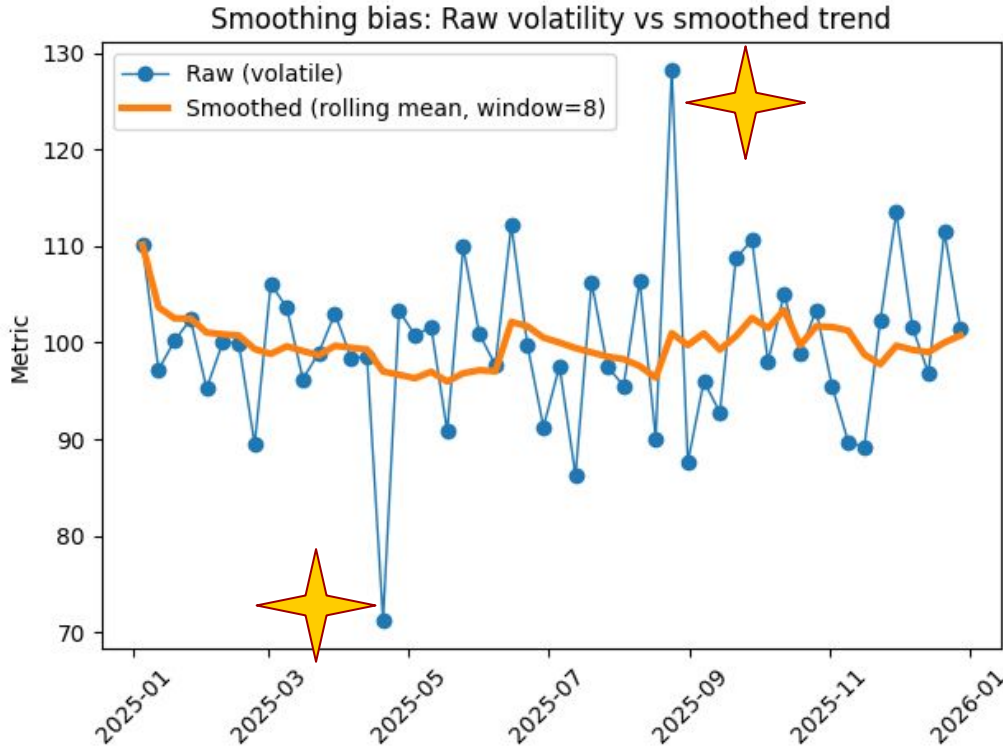
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Max window

Zooming too far in (or out) can mislead trends.

Bias: Smoothing

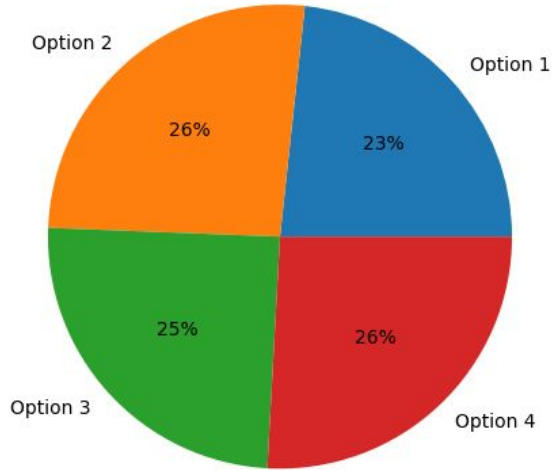


- Rolling window averages (orange) can hide the volatility of raw data.
- Useful for extracting long term trends in noisy data but be wary of abuse/misuse/obfuscation.

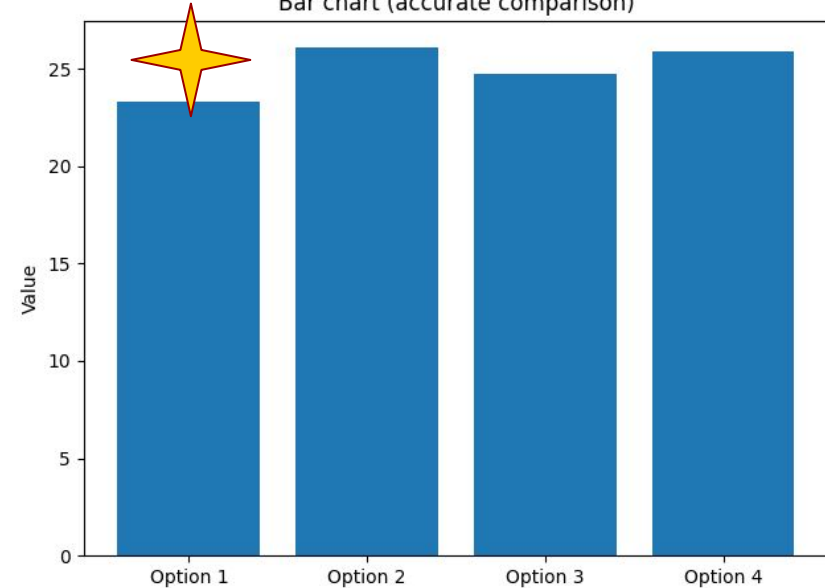
Bias: Chart Type Misuse



Pie chart (misleading for close values)



Bar chart (accurate comparison)



Intentional or not, a pie chart can obscure accuracy/precision of close call results clearly visible in a bar chart.



Other biases

Color: Using **bright green** or **bright red** to exaggerate and project success or failure when the raw data/numbers don't support it.

Aggregation: Using mean/average, median, mode, min, max, or sum/totals instead of raw numbers to hide poor, skewed, or favoring distributions.

Framing: Using sensationalist wording (usually along with other unethical techniques) e.g. Titles like “**Skyrocketing** sales” or “**Collapse** of” used along w/ selection, framing, color, and aggregation biases when the raw data numbers were only a few percent differences.

Poll: Has anyone seen these biases lately?



- Newspapers
- Social
- TV/Movies
- White papers
- Journals
- etc.

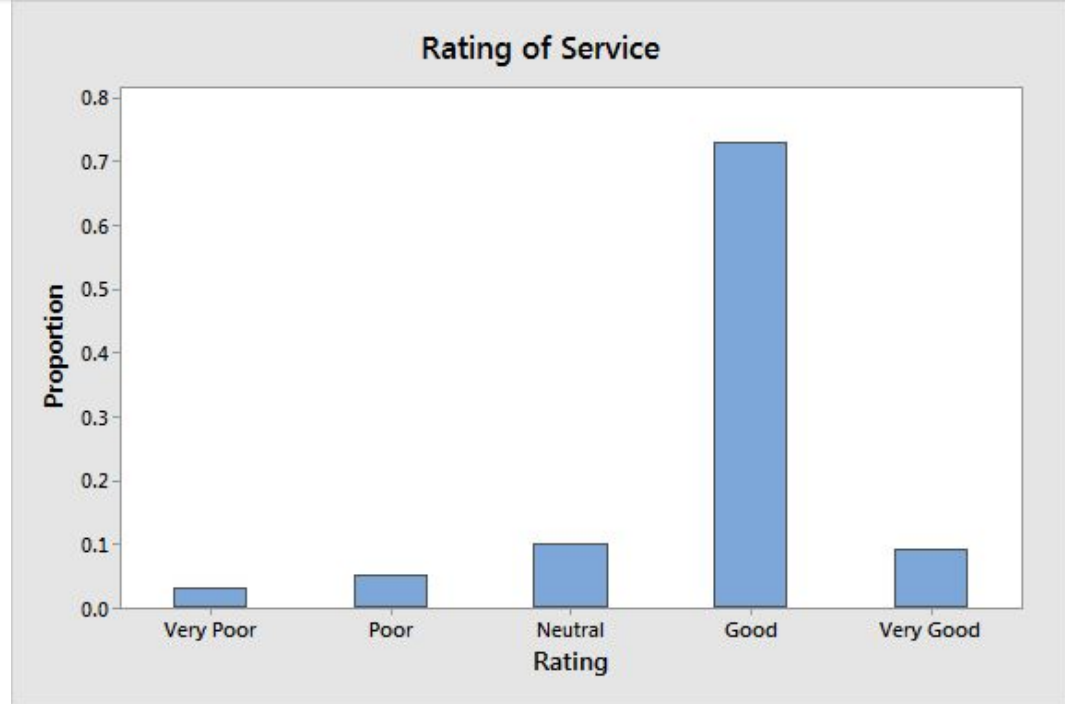


Visual literacy for executives, regulators, and the public

Visual literacy for the public



- **What to focus on:** simplicity and clarity, titles, labels, units, infographics
- **Best chart type:** simple bars and lines
- **Avoid:** overwhelming with data, pie charts, framing bias, forcing eyes to move



<https://statisticsbyjim.com/basics/ordinal-data/>

Visual literacy for regulators



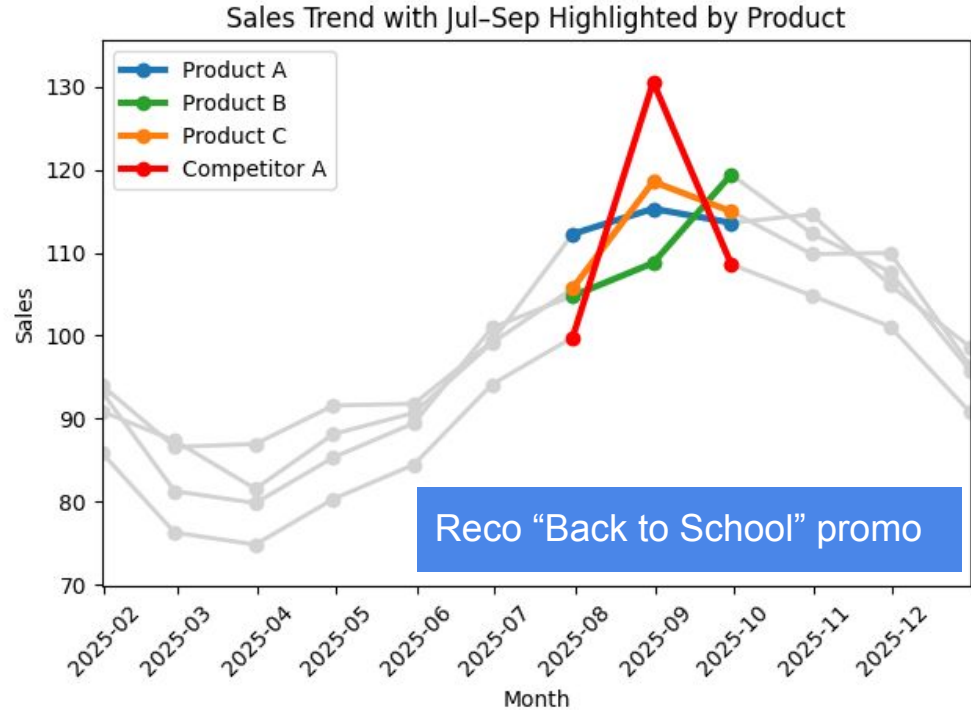
- **What to focus on:** precision, detail, accuracy, fixed axes for comparison, numerical analysis as evidence, dashboards
- **Best chart type:** annotated, fixed axes line plot, histogram, box plot, heatmap
- **Avoid:** aggregations that can hide sub-groups, scale, selection, framing biases



Visual literacy for executives



- **What to focus on:** overall, broad, trends, boombox, aggregations, rolling window averages, etc., dashboards, KPIs, BLUF, Ask
- **Best chart type:** highlighted line, bar, geospatial, choropleth
- **Avoid:** clutter, reporting instead of storytelling



Review HW1



- Go over instructions in-class
- Questions about HW?
- Reminder:
 - Piazza, Piazza, Piazza
 - Office Hours (instructor and/or TA)
 - Open Lab Times (either LA)

Questions?

